

Applied Time Series Analysis

Introduction to Time Series Analysis and Basic Concepts

Objectives:

- What is time series data?
- How time series differs from other data types
- Applications of time series analysis
- Components of time series
- Time series notation
- Trend, seasonality, cyclicity, and irregular variations
- Time series forecasting workflow
- Basic concepts of stationarity and autocorrelation

Introduction to Time Series Analysis

What is Time Series Data?

A **time series** is a sequence of observations collected on the same variable at different points in time.

Mathematically:

$Y_1, Y_2, Y_3, \dots, Y_t$

where:

- Y = observed value
- t = time index

Example:

Monthly airline passengers

Month	Passengers
Jan 1949	112
Feb 1949	118
Mar 1949	132
Apr 1949	129

- Variable = Number of passengers
- Time = Month
- Observation = Passenger count

The order of observations is important.

Difference Between Time Series Data and Ordinary Data

1. Cross-sectional Data

Data collected at one point in time.

Example:

Student marks:

Student	Marks
A	85
B	90
C	78

The order does not matter.

2. Time Series Data

Data collected over time.

Example:

Monthly sales:

Month	Sales
January	10000
February	12000
March	15000

The order matters.

January affects February, February affects March.

This dependency is called Temporal dependence

Applications of Time Series Analysis

Time series analysis is used whenever future values need to be predicted.

1. Business Forecasting

Example:

A company wants to predict next month's sales.

Data:

January sales
February sales
March sales

Uses:

- Inventory planning
- Production planning
- Revenue forecasting

2. Financial Markets

Examples:

- Stock prices
- Exchange rates
- Interest rates

Questions:

- What will be tomorrow's stock price?
- What is the market trend?

3. Weather Forecasting

Data:

- Temperature
- Rainfall
- Wind speed

Example:

Past temperature:

$30^{\circ}\text{C} \rightarrow 32^{\circ}\text{C} \rightarrow 31^{\circ}\text{C} \rightarrow 33^{\circ}\text{C}$

Predict future temperature.

4. Healthcare

Examples:

- Patient monitoring
- Disease spread
- Hospital admissions

5. Energy Forecasting

Examples:

- Electricity consumption
- Gas demand

Used for:

- Power generation planning
- Grid management

Characteristics of Time Series Data

A time series has special properties.

1. Time Order Matters

Example:

Monthly sales:

January: 100

February: 150

March: 200

Changing order:

March: 200

January: 100

February: 150

changes the meaning.

2. Observations are Dependent

In normal statistics, observations may be independent.

Example:

Height of different people:

Person A height does not affect Person B.

But in time series:

Today's temperature depends partly on yesterday's temperature.

This relationship is called:

Autocorrelation

Meaning:

"Correlation of a variable with its past values."

Example:

Y_t depends on Y_{t-1}

Where Y_t = today's value and Y_{t-1} = previous value

3. Data Frequency

Frequency means how often observations are recorded.

Examples:

Yearly

GDP:

2010

2011

2012

Monthly

Sales:

Jan

Feb

Mar

Daily

Temperature:

1 Jan

2 Jan

3 Jan

Hourly

Electricity usage:

10 AM

11 AM

12 PM

Components of Time Series

A time series is usually divided into four components:

$$Y_t = T_t + S_t + C_t + E_t$$

where:

- T_t = Trend
- S_t = Seasonal
- C_t = Cyclic
- E_t = Error/Random

1. Trend Component

Trend represents the long-term movement of data.

It can be:

Increasing Trend

Example:

Company sales:

2019	1000
2020	1500
2021	2000
2022	2500

Sales are increasing.

Decreasing Trend

Example:

Newspaper sales:

2019	50000
2020	40000
2021	30000

No Trend

Values fluctuate around a constant average.

2. Seasonal Component

Seasonality is a pattern that repeats at fixed intervals.

Examples:

Ice cream sales

Summer:

High sales

Winter:

Low sales

Pattern repeats every year.

Monthly retail sales

During festivals:

Higher sales

After festivals:

Lower sales

Seasonality has a fixed period.

Examples:

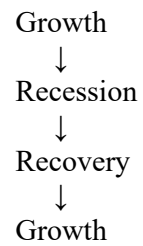
- Daily data → weekly pattern
- Monthly data → yearly pattern
- Hourly data → daily pattern

3. Cyclic Component

Cycles are long-term fluctuations without a fixed period.

Example:

Economic cycles:



Difference:

Seasonal	Cyclic
Fixed period	No fixed period
Short duration	Long duration

4. Random Component

Random variation is unpredictable movement.

Examples:

- Natural disasters
- Sudden market crashes
- Unexpected events

Represented as E_t

Time Series Decomposition

Explain:

Observed data contains all components together.

Example:

Air passenger data:

Observed:

Passengers increasing every year

After decomposition:

Trend

Shows long-term growth

Seasonal

Shows yearly travel patterns

Residual

Shows random changes

Two decomposition models:

Additive Model

Used when seasonal variation is constant.

Formula:

$$Y_t = T_t + S_t + E_t$$

Example:

Seasonal increase always around +100 passengers.

Multiplicative Model

Used when seasonal variation increases with trend.

Formula:

$$Y_t = T_t \times S_t \times E_t$$

Example:

Air passenger data.

As passenger numbers increase, seasonal changes also become larger.

Stationarity Concept

One of the most important concepts.

What is Stationarity?

A stationary time series has statistical properties that do not change over time.

These properties include:

1. Constant Mean

Average remains stable.

Example:

10
12
11
9
10

2. Constant Variance

Spread remains stable.

3. Constant Relationship Over Time

Past behavior continues similarly.

Non-Stationary Example

Stock price:

100
120
150
180
220

Mean changes over time.

This is non-stationary.

Why stationarity matters?

Many forecasting models assume stationary data.

Examples:

- AR model
- MA model
- ARIMA model

If data is not stationary, we may transform it.

Methods:

- Differencing
- Log transformation

Forecasting Workflow

Explain the complete process:

Step 1: Data Collection

Example:

Collect monthly sales data.

Step 2: Data Visualization

Plot data.

Look for:

- Trend
- Seasonality
- Outliers

Step 3: Data Preparation

Tasks:

- Handle missing values
- Convert dates
- Remove errors

Step 4: Analyze Time Series

Methods:

- Decomposition
- Stationarity tests
- Autocorrelation analysis

Step 5: Build Forecasting Model

Examples:

Beginner:

- Moving Average
- Exponential Smoothing

Advanced:

- ARIMA
- SARIMA
- Machine Learning
- Deep Learning

Step 6: Evaluate Forecast

Compare:

Actual values

vs

Predicted values

Metrics:

- MAE
- RMSE
- MAPE